

Column 2 Sensitivity Analysis Report

Objective

The objective of this study was to investigate the influence of the major operating parameters of the solvent recovery column (Column 2) on process performance and identify the variables that significantly affect ethanol recovery, ethylene glycol recovery, product purity, energy consumption, and column convergence.

The operating variables investigated were:

- Bottom Product Flow Rate
- Reflux Ratio
- Feed Stage
- Number of Stages
- Column Pressure

Each parameter was varied independently while maintaining all other operating conditions constant.

Base Case

The sensitivity analysis was performed using the validated operating conditions obtained from Column2-

Feed Flow - 21918.7 kg/h	Feed Stage - 8
Number of Stages - 15	Feed Temperature - 179.652 °C
Feed Pressure - 3.5 bar	Column Pressure - 1 bar
Reflux Ratio - 1.5	Bottom Flow - 0.09 kmol/s

Performance Indicators

The following responses were monitored for every simulation.

- Ethanol Recovery
- Ethylene Glycol Recovery
- Ethanol Purity (Top Product)

- Total Energy Consumption
- Column Convergence

Only fully converged simulations were considered for analysis.

1. Bottom Flow Sensitivity

Bottom product flow rate was varied between 0.07 kmol/s and 0.13 kmol/s.

Observations

- Increasing bottom flow significantly increased ethanol purity.
- EG recovery also increased and approached complete recovery.
- Total energy consumption decreased considerably.
- Extremely high bottom flow values (>0.10 kmol/s) resulted in non-converged simulations.

Engineering Interpretation

Bottom flow determines the quantity of solvent retained within the recovery column.

A higher bottom withdrawal removes ethylene glycol more efficiently while reducing ethanol losses and decreasing the overall thermal duty required for separation.

Among all investigated parameters, bottom flow exhibited the strongest influence on overall column performance.

Conclusion

Bottom Flow is the most critical operating variable for Column 2 and was therefore selected as one of the primary optimization variables.

2. Reflux Ratio Sensitivity

The reflux ratio was varied from 0.01 to 2.00.

Observations

- Ethanol recovery remained almost unchanged.
- Ethylene glycol recovery remained nearly constant after $RR \approx 0.9$.
- Ethanol purity improved only marginally.

- Energy consumption increased almost linearly with increasing reflux ratio.

Engineering Interpretation

Increasing reflux beyond the minimum required value does not significantly improve mass transfer because the solvent recovery column is already operating close to equilibrium. Additional reflux mainly increases condenser and reboiler duties.

Conclusion

A reflux ratio of approximately 1.0 provides the best compromise between separation efficiency and energy consumption. Increasing reflux beyond this value is not economically justified.

3. Feed Stage Sensitivity

Feed location was varied from Stage 2 to Stage 11.

Observations

- Feed Stage 2 produced noticeably lower ethanol purity.
- Moving the feed to Stage 8 significantly improved product quality.
- Feed stages above 8 produced almost identical results.
- Energy consumption showed only minor variation.

Engineering Interpretation

The feed should be introduced near the region where internal liquid and vapor compositions closely match the feed composition. Beyond the optimum feed location, additional changes provide negligible improvement.

Conclusion

Stage 8 represents the optimum feed location and was selected for subsequent optimization studies.

4. Number of Stages Sensitivity

Theoretical stages were varied between 8 and 16.

Observations

- Ethanol recovery remained almost constant.
- EG recovery remained almost constant.
- Ethanol purity changed negligibly.
- Energy consumption remained essentially unchanged.

Engineering Interpretation

The solvent recovery column reaches equilibrium within relatively few stages. Adding extra trays increases column height and capital cost without improving separation.

Conclusion

Increasing the number of stages beyond 10 provides negligible benefit. Additional trays are therefore economically unjustified.

5. Column Pressure Sensitivity

Column pressure was varied between 0.8 bar and 3.5 bar.

Observations

- Product purity changed very little.
- Recovery remained almost constant.
- Energy consumption increased steadily with pressure.
- Temperature also increased with pressure.

Engineering Interpretation

Higher operating pressure increases boiling temperature and consequently raises reboiler duty. Since separation performance remained nearly unchanged, higher pressure only increased operating cost.

Conclusion

Atmospheric pressure (1 bar) provides the best balance between separation efficiency and energy consumption.

Overall Parameter Importance

The influence of each operating parameter is summarized below.

Parameter	Effect on Separation	Effect on Energy	Overall Importance
Bottom Flow	Very High	Very High	☆☆☆☆☆
Reflux Ratio	Low	Very High	☆☆☆☆
Feed Stage	Moderate	Low	☆☆☆
Column Pressure	Low	Moderate	☆☆
Number of Stages	Negligible	Negligible	☆

Major Findings

The sensitivity analysis demonstrated that:

- Bottom Flow is the dominant operating parameter governing solvent recovery.
- Reflux ratio primarily influences energy consumption rather than separation performance.
- Feed Stage 8 provides the optimum feed location.
- Increasing the number of stages beyond 10 offers no meaningful improvement.
- Atmospheric pressure minimizes energy consumption while maintaining high recovery.

Final Design Decisions

Based on the sensitivity analysis, the following operating conditions were adopted for subsequent validation and optimization studies.

Reason	Parameter	Selected Value
Highest recovery with stable convergence	Bottom Flow	0.09 kmol/s
Minimum energy penalty	Reflux Ratio	1.0
Highest separation efficiency	Feed Stage	8
Additional stages unnecessary	Number of Stages	10
Lowest energy consumption	Column Pressure	1 bar

Summary

The sensitivity analysis successfully identified the operating variables that significantly affect Column 2 performance. The study demonstrated that bottom product flow is the primary design variable, whereas reflux ratio, feed stage, and pressure mainly influence process economics rather than separation efficiency. These findings formed the engineering basis for the validation study and the subsequent optimization of the complete extractive distillation process.